

Restaurant and Review Analysis Report

Restaurant Analytics Using the Zomato Dataset

Submitted By

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1. Introduction

Online restaurant discovery platforms generate large amounts of data about restaurants, customer preferences, ratings, pricing, and cuisines. Analyzing this data helps businesses understand customer behavior and improve their services. This project analyzes the Zomato restaurant dataset to identify important trends related to ratings, cuisines, locations, and pricing. The results provide useful business insights and recommendations for an Alfido Tech-style restaurant platform.

2. Project Objectives

The objectives of this project are:

- Clean and preprocess the restaurant dataset.
 - Handle missing values and text fields.
 - Analyze restaurant ratings and customer preferences.
 - Identify the most popular cuisines.
 - Find restaurant hotspots based on location.
 - Study the relationship between price and ratings.
 - Create meaningful visualizations.
 - Provide business recommendations for Alfido Tech.
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3. Dataset Overview

The dataset contains approximately **56,000 restaurant records** with information such as:

- Restaurant Name
- Online Order Availability
- Table Booking
- Customer Rating
- Number of Votes
- Restaurant Location
- Restaurant Type
- Cuisines
- Approximate Cost for Two People
- Reviews
- Menu Items
- Listed Category

This information is useful for understanding restaurant popularity and customer satisfaction.

4. Data Cleaning

The following preprocessing steps were performed:

- Removed duplicate records.
- Checked for missing values.
- Filled missing values using suitable techniques.
- Converted ratings from text to numeric values.
- Removed commas from cost values.
- Converted approximate cost into numeric format.
- Cleaned cuisine and restaurant type columns.
- Standardized text data for analysis.

These preprocessing steps improved data quality and ensured accurate analysis.

5. Exploratory Data Analysis

The following analyses were carried out:

Rating Distribution

Most restaurants have ratings between **3.5 and 4.2**, indicating generally good customer satisfaction.

Popular Cuisines

The most common cuisines are:

- North Indian
- Chinese
- South Indian
- Biryani
- Fast Food

These cuisines dominate the restaurant market.

Restaurant Hotspots

Restaurants are mainly concentrated in commercial locations such as:

- BTM
- Koramangala
- HSR Layout
- Indiranagar
- Whitefield

These areas have high restaurant density and customer demand.

Price vs Rating

Higher-priced restaurants tend to receive slightly higher ratings. However, many affordable restaurants also achieve excellent ratings.

Online Ordering

Restaurants that provide online ordering generally receive better customer engagement and higher ratings.

Table Booking

Restaurants offering table booking facilities usually maintain better customer ratings because of improved convenience.

Votes vs Ratings

Restaurants with higher customer votes generally have higher ratings, indicating strong customer trust and popularity.

6. Visualizations

The notebook includes the following visualizations:

- Missing Value Heatmap
- Rating Distribution Histogram
- Top Restaurant Locations Bar Chart
- Popular Cuisine Bar Chart
- Cuisine vs Rating Box Plot
- Price vs Rating Scatter Plot
- Online Order vs Rating Box Plot
- Table Booking vs Rating Box Plot
- Correlation Heatmap
- Restaurant Type Distribution
- Votes vs Rating Scatter Plot
- Popular Cuisine Word Cloud

These visualizations clearly explain the restaurant trends and customer preferences.

7. Key Findings

- Most restaurants receive ratings between **3.5 and 4.2**.
 - North Indian and Chinese cuisines are the most preferred.
 - Commercial areas have the highest restaurant density.
 - Online ordering contributes to better customer engagement.
 - Restaurants offering table booking receive slightly higher ratings.
 - Customer votes have a positive relationship with restaurant ratings.
 - Restaurant price has only a moderate impact on customer ratings.
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8. Recommendations for Alfido Tech

1. Partner with Highly Rated Restaurants

Collaborate with restaurants having ratings above 4.0 to improve customer trust and platform quality.

2. Promote Popular Cuisines

Highlight trending cuisines like North Indian, Chinese, and Biryani through featured collections.

3. Personalized Recommendations

Recommend restaurants based on customer location, preferred cuisines, and budget.

4. Encourage Customer Reviews

Provide reward points, coupons, or loyalty benefits to customers who submit genuine reviews.

5. Location-Based Marketing

Run promotional campaigns targeting restaurant hotspot areas to attract more users and increase engagement.

9. Conclusion

The Restaurant and Review Analysis project successfully analyzed restaurant data using Python and data visualization techniques. The study identified the major factors affecting restaurant ratings, including cuisine type, location, online ordering, and customer engagement. The generated insights can help Alfido Tech improve restaurant recommendations, customer satisfaction, and marketing strategies. Overall, this project demonstrates how data analytics can support better business decision-making in the food delivery industry.

10. Tools and Technologies Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - WordCloud
 - Jupyter Notebook
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Deliverables

- ✓ Cleaned Dataset
 - ✓ Jupyter Notebook with Data Cleaning
 - ✓ Exploratory Data Analysis
 - ✓ Visualizations
 - ✓ Business Insights
 - ✓ PDF Report
 - ✓ Five Recommendations for Alfido Tech
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